

Support Vector Machines (SVMs) use **kernels** to handle non-linear relationships by implicitly mapping data into higher-dimensional spaces. Choosing the right kernel can have a big impact on performance. Here are the most common SVM kernels:

1. Linear Kernel ([Read More](#))

- Pros
 - Fast and memory-efficient
 - Less risk of overfitting
 - Works well with many features
- Cons
 - Cannot capture complex, non-linear patterns

2. Polynomial Kernel ([Read More](#))

- Pros
 - Can model interactions between features
 - More flexible than linear
- Cons
 - Higher degrees can overfit
 - Computationally more expensive

3. RBF (Gaussian) Kernel ([Read More](#))

- Pros
 - Very flexible
 - Works well for many real-world problems
- Cons
 - Sensitive to hyperparameters
 - Can overfit if γ is too large

4. Sigmoid Kernel ([Read More](#))

- Pros
 - Can mimic neural network decision boundaries
- Cons
 - Not always a valid kernel for all parameters
 - Often performs worse than RBF

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